Why 96% of Customers
Did Not Repurchase Within 180 Days

A Customer Retention Strategy for Olist Fashion

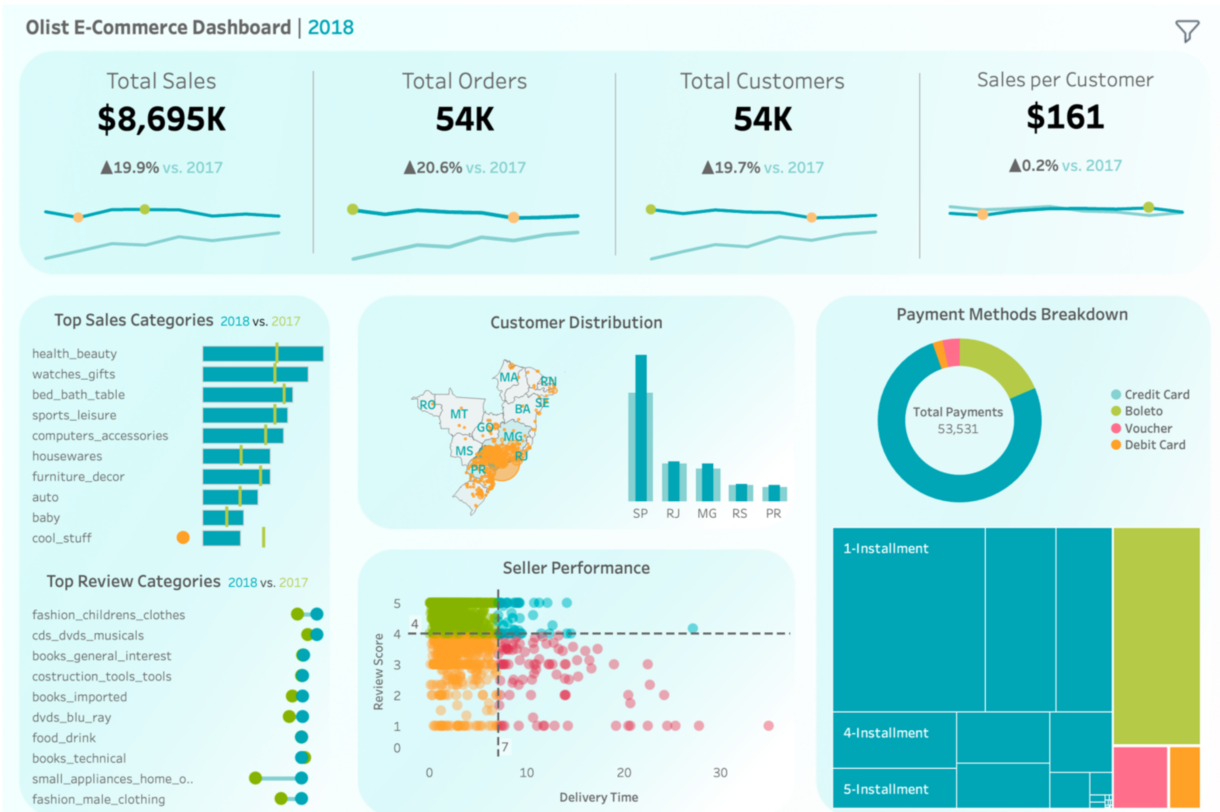
Executive Summary

Business Problem	• Olist grew strongly in 2018, reaching $8.7M in sales, 54K orders, and 54K customers, but sales, orders, and customer growth began to stabilize. • Only 3.61% of Olist Fashion customers made a second purchase within 180 days, meaning 96.39% did not repurchase within the six-month analysis window.
Analytical Approach	• Customer Segmentation - RFM-based K-Means clustering identified One-Timers, Newcomers, Repeat Customers, and High-Value VIPs. • Churn Driver Analysis - Decision Tree and segmented Logistic Regression tested the impact of order value, freight cost, and delivery delay on repurchase behavior. • CRM Strategy Design - A fulfillment-latency view translated delivery risk into early-warning CRM actions.
Key Findings	• A $98.71 order-value split separated Value customers from Premium customers. • Value customers are more sensitive to product price and shipping cost; Premium customers are more sensitive to delivery delays. • NLP review mining further supported fulfillment experience as a key retention-risk signal.
Strategic Recommendation	• Olist should move from one-size-fits-all retention management to segment-specific CRM strategy. • Value customers need shipping incentives and repeat-purchase activation; Premium customers need fulfillment reliability, delivery transparency, and proactive service recovery.

1. Business Context

Marketplace Growth Overview

An executive Tableau dashboard was developed to assess Olist's marketplace performance across revenue, orders, customer growth, geographic distribution, category performance, seller performance, and payment behaviour.



The interactive dashboard can be found [here](#).

Key Finding

Olist achieved strong year-over-year growth in 2018, but growth across sales, orders, and customer count began to stabilize.

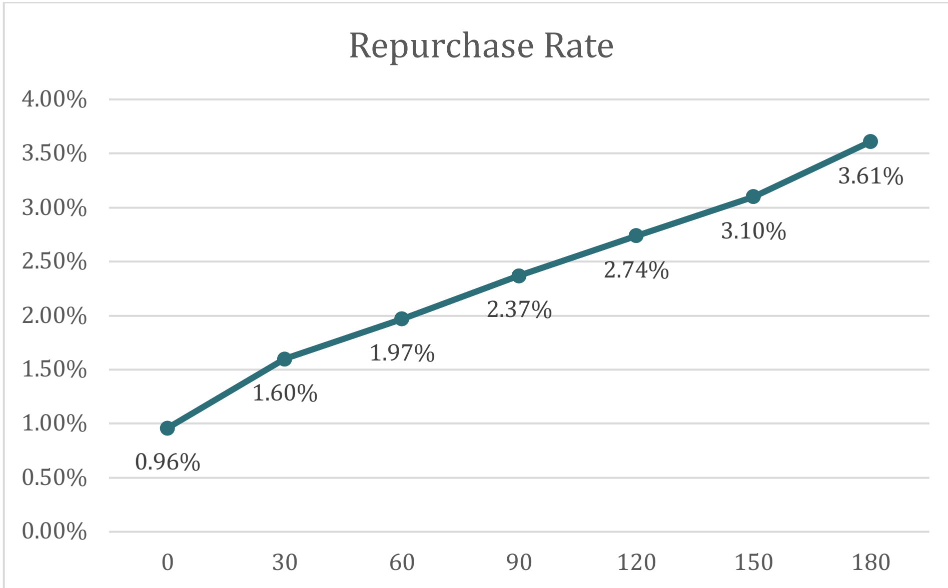
Business Interpretation

This pattern suggests that Olist may be facing a retention-quality problem beneath headline growth. If growth is mainly driven by new customer acquisition rather than repeat purchasing, long-term marketplace performance may become harder to sustain.

2. Diagnosing the Retention Problem

Measuring Repurchase Rate

To evaluate customer retention, cumulative repurchase rates were calculated after each customer's first purchase.



Results

- 30 days: 1.60%
- 90 days: 2.37%
- 180 days: 3.61%

Only 3.61% of Olist Fashion customers made a second purchase within 180 days.

Benchmark Context

Although repeat-purchase benchmarks vary by category, business model, and measurement window, Olist Fashion's 3.61% 180-day repurchase rate appears materially low compared with public ecommerce benchmarks. Klaviyo notes that a good repeat purchase rate is typically around 20%–30%, while a DTC benchmark across 156,110 customers and multiple brands reported an average 18.8% repeat purchase rate within a 365-day window (Klaviyo; B&S&Co.). These benchmarks are not direct one-to-one comparisons because Olist's rate is measured over 180 days and focuses on the Fashion category, but they provide directional evidence that Olist Fashion faces a meaningful retention gap.

Business Interpretation

This suggests a classic "leaky bucket" problem: Olist continues acquiring customers, but very few remain active long enough to generate repeat revenue.

Customer retention therefore represents a major growth opportunity. Improving post-purchase engagement and repeat purchase behaviour may create stronger long-term value than relying only on customer acquisition.

3. Customer Segmentation: Who Are the Customers?

RFM-Based K-Means Clustering

To understand customer behaviour, RFM features were created for Fashion customers:

- Recency
- Frequency

- Monetary value

After preprocessing and scaling, K-Means clustering was applied to identify four behavioural customer personas.

Cluster ID	Segment	User Count	Avg. Recency	Avg. Frequency	Avg. Monetary
0	One-Timers	34,719	394.0	1.05	$110.79
1	Newcomers	46,938	129.8	1.00	$108.46
2	Repeat Customers	9,655	193.3	2.57	$199.54
3	High-Value VIPs	2,046	235.8	1.27	$1180.65

Results

Cluster 0 - One-Timers

Single-purchase customers with minimal engagement.

Cluster 1 - Newcomers

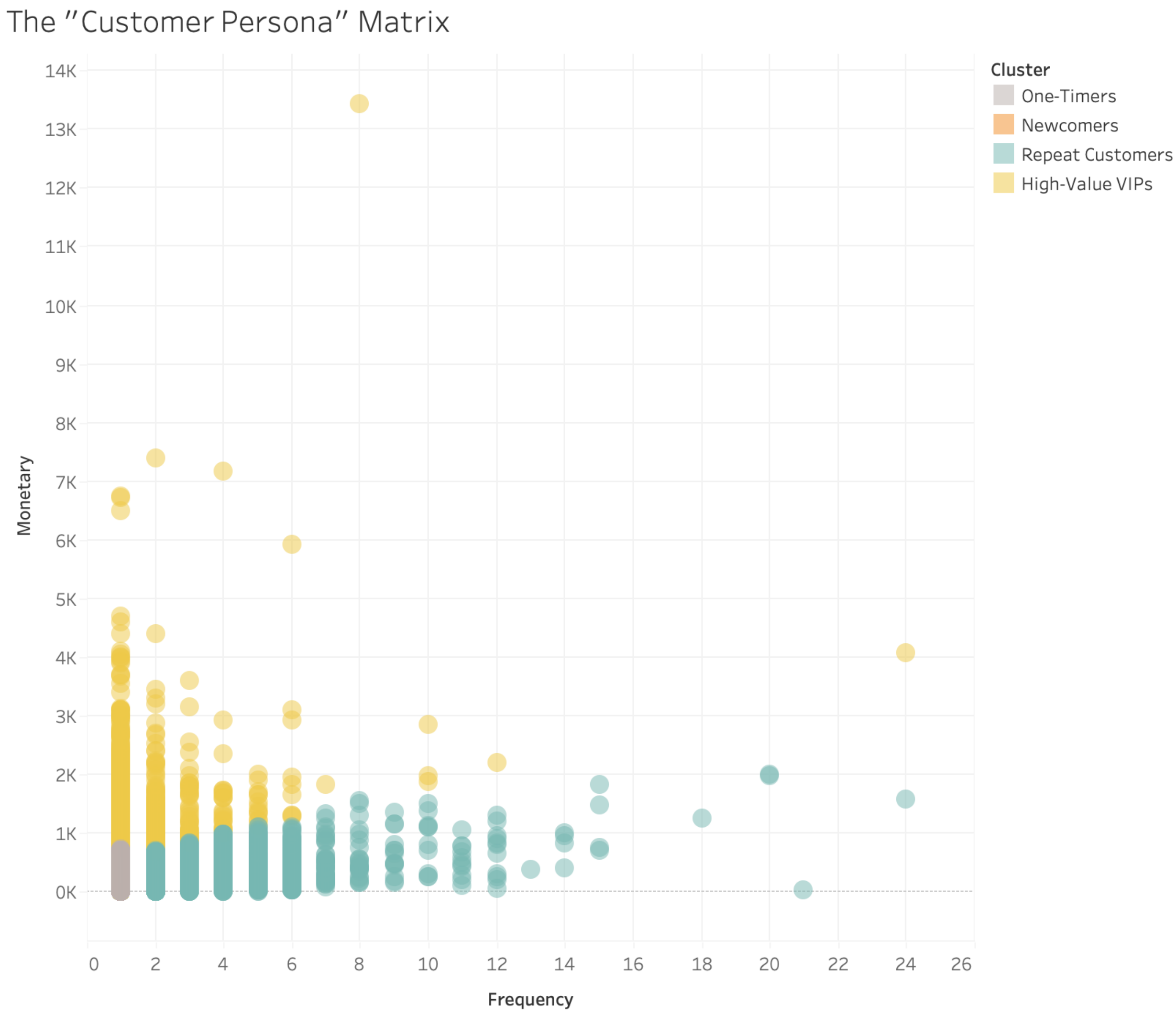
Recently acquired customers with limited purchasing history.

Cluster 2 - Repeat Customers

Frequent buyers with relatively low spending per order.

Cluster 3 - High-Value VIPs

Customers with the highest spending and strongest revenue contribution.



Business Interpretation

The segmentation shows that Olist Fashion customers differ meaningfully in purchase frequency and monetary value. However, high recency across clusters suggests that retention issues are not limited to one customer type.

This section answers who the customers are. The next section tests why they fail to repurchase.

4. Hypothesis Testing: What Drives Repurchase?

4.1 Initial Hypothesis

To understand why most Fashion customers did not repurchase, two operational hypotheses were tested:

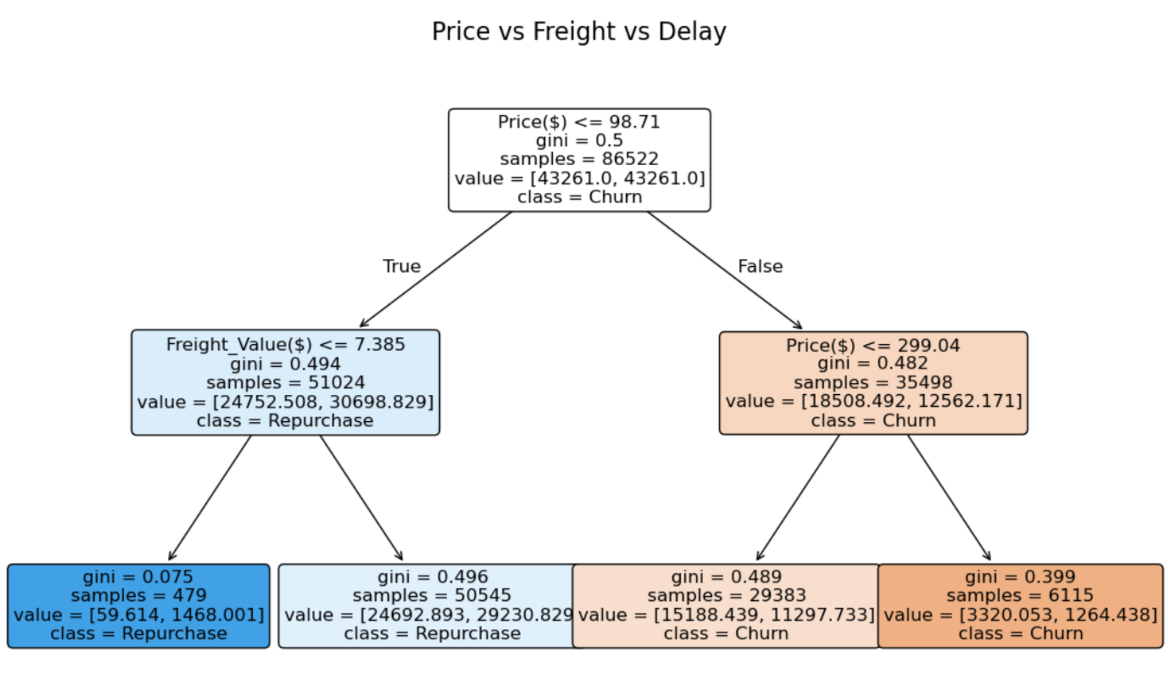
1. Delivery delays may reduce repurchase likelihood.
2. Shipping cost may create purchase friction, especially for lower-value orders.

A two-step testing approach was used.

First, a Decision Tree was applied to identify major behavioural split points in the customer base. Second, segmented Logistic Regression was used to quantify how product price, freight cost, and delivery delay influenced repurchase behaviour within each segment.

4.2 First Test: Decision Tree Analysis

A Decision Tree was used to identify the strongest behavioural split points related to repurchase behaviour.



Result

The Decision Tree identified average order value = $98.71 as the first major behavioural split.

This threshold separated customers into two economically different groups:

- Value Segment: Order Value < $98.71
- Premium Segment: Order Value >= $98.71

Within the Value Segment, freight cost also emerged as an important split: Freight Cost < $7.38.

Business Interpretation

The Decision Tree suggests that order value is the strongest behavioural divider. Among lower-order-value customers, freight cost below $7.38 is associated with a higher likelihood of repurchase.

This finding reframes the shipping-cost hypothesis. Freight cost does not affect all customers equally. Instead, it appears most relevant for lower-order-value customers, where shipping cost may feel more visible relative to the total order value.

4.3 From Decision Tree to Segmented Analysis

The Decision Tree result became the foundation for the next stage of analysis.

Instead of modeling all customers as one group, customers were divided into two segments based on the $98.71 threshold:

- Value Segment: customers with order value below $98.71
- Premium Segment: customers with order value at or above $98.71

This step is important because the Decision Tree suggested that customers do not respond to operational factors in the same way. Therefore, a single overall Logistic Regression model could hide important differences between value-oriented and premium customers.

In short: Decision Tree defines the segments. Logistic Regression explains the drivers within each segment.

4.4 Second Test: Segmented Logistic Regression

Separate Logistic Regression models were built for the Value Segment and Premium Segment.

The dependent variable was whether a customer repurchased within the analysis window. The explanatory variables included:

- Product price
- Freight cost
- Delivery delay

Because customer repurchase is influenced by many factors beyond the available dataset, including customer preferences, marketing exposure, product satisfaction, and competitive alternatives, the models are used for directional diagnosis rather than individual-level prediction.

The goal is not to build a high-accuracy prediction model, but to compare whether the same operational factors affect value and premium customers differently.

4.4.1 Value Segment Findings

A segmented Logistic Regression model was built to measure how product price, freight cost, and delivery delay influenced repurchase behavior among value customers.

```
                           Logit Regression Results
==============================================================================
Dep. Variable:         is_repurchased   No. Observations:                51024
Model:                          Logit   Df Residuals:                    51020
Method:                           MLE   Df Model:                            3
Date:                Sat, 30 May 2026   Pseudo R-squ.:                0.001975
Time:                        13:28:01   Log-Likelihood:                -21849.
converged:                       True   LL-Null:                       -21892.
Covariance Type:            nonrobust   LLR p-value:                 1.249e-18
=================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const            -1.4351      0.035    -40.646      0.000      -1.504      -1.366
price            -0.0042      0.001     -7.822      0.000      -0.005      -0.003
freight_value    -0.0037      0.002     -2.142      0.032      -0.007      -0.000
is_delayed       -0.1469      0.053     -2.758      0.006      -0.251      -0.043
=================================================================================
```

Result

For value customers, the model showed that:

- Higher product price is associated with lower repurchase likelihood.
- Higher freight cost is associated with lower repurchase likelihood.
- Delivery delays reduce repurchase likelihood.

Business Interpretation

Value customers are sensitive to both affordability and fulfillment experience. For this group, retention strategies should focus on reducing purchase friction through shipping incentives, bundled promotions, and first repeat-purchase offers.

4.4.2 Premium Segment Findings

A segmented Logistic Regression model was built to measure how the same factors influenced repurchase behaviour among premium customers.

```
                           Logit Regression Results
==============================================================================
Dep. Variable:          is_repurchased   No. Observations:                35498
Model:                           Logit   Df Residuals:                    35494
Method:                            MLE   Df Model:                            3
Date:                 Sat, 30 May 2026   Pseudo R-squ.:                0.007020
Time:                         13:28:02   Log-Likelihood:                -10697.
converged:                        True   LL-Null:                       -10772.
Covariance Type:             nonrobust   LLR p-value:                 1.417e-32
=================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const            -2.0916      0.035    -60.162      0.000      -2.160      -2.023
price            -0.0012      0.000     -9.483      0.000      -0.001      -0.001
freight_value     0.0025      0.001      2.670      0.008       0.001       0.004
is_delayed       -0.4026      0.081     -4.942      0.000      -0.562      -0.243
=================================================================================
```

Result

For premium customers, the model showed that:

- Product price has limited influence.
- Freight cost has limited influence.
- Delivery delay becomes the strongest churn-related factor.

Business Interpretation

Premium customers are less discount-driven. Their repurchase behaviour is more closely linked to fulfillment reliability and service quality.

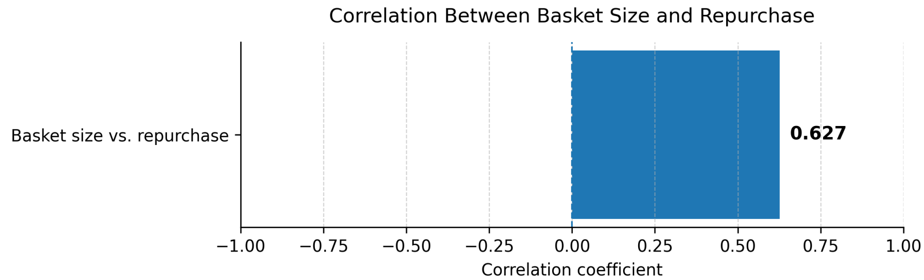
This means that premium customer retention should focus less on price incentives and more on delivery transparency, priority fulfillment, and proactive service recovery.

5. Testing an Alternative Explanation

Are High-Value Customers Simply Stockpiling?

One alternative explanation was that high-value customers buy in larger quantities and therefore naturally take longer to repurchase.

To test this hypothesis, the relationship between basket size and repurchase behavior was measured.



Result

Basket size showed a positive relationship with repurchase. Customers who purchased more items were actually more likely to return.

Business Interpretation

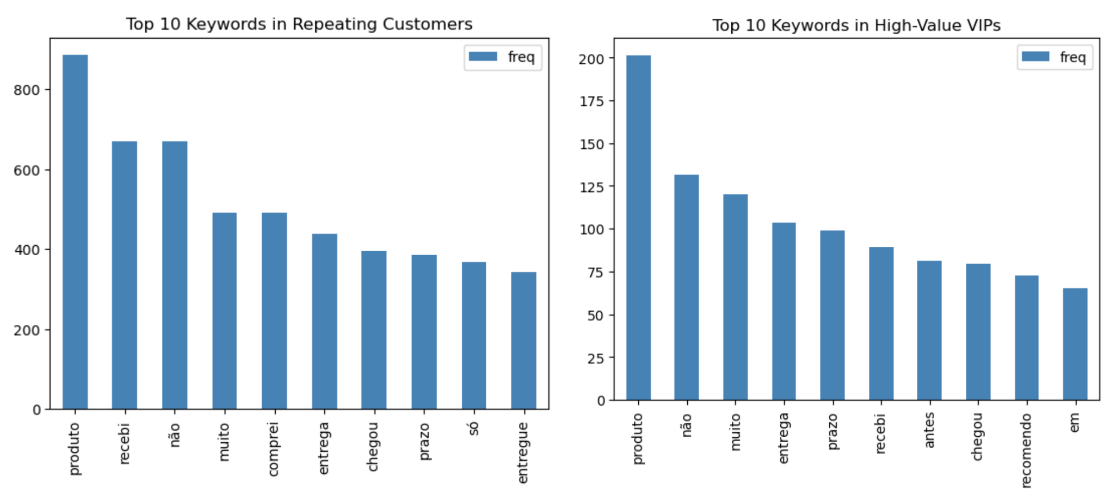
This weakens the stockpiling explanation. If high-value customers were simply buying in bulk and delaying their next purchase, larger basket size would likely be associated with lower repurchase behavior. Instead, larger baskets were associated with stronger repeat behavior.

High-value customer churn is therefore unlikely to be explained by bulk-purchase behavior alone. Service experience and delivery performance remain more plausible explanations for premium customer churn.

6. Voice of Customer Analysis

NLP Review Mining

To understand customer sentiment, review text was analyzed for Repeat Customers and High-Value VIPs. Keyword extraction and text preprocessing were used to identify recurring themes in customer feedback.



Portuguese Keyword	English Meaning
produto	Product
recebi	Received
não	No / Not
muito	Very

comprei	Bought
entrega	Delivery
prazo	Deadline / Term
chegou	Arrived
entregue	Delivered
antes	Before
recomendo	Recommend

Result

Delivery-related terms appeared frequently across both segments. Common themes included delivery delays, shipping experience, and order arrival timing.

Common Portuguese keywords included:

- entrega - delivery
- prazo - deadline / delivery time
- chegou - arrived
- recebi - received

High-Value VIPs also frequently used recommendation-oriented language, such as “recomendo.”

Business Interpretation

The review analysis supports fulfillment experience as a key retention-risk signal. High-Value VIPs represent both revenue value and referral value. Retaining this group can therefore create a disproportionately large business impact.

7. Fulfillment Early-Warning Framework

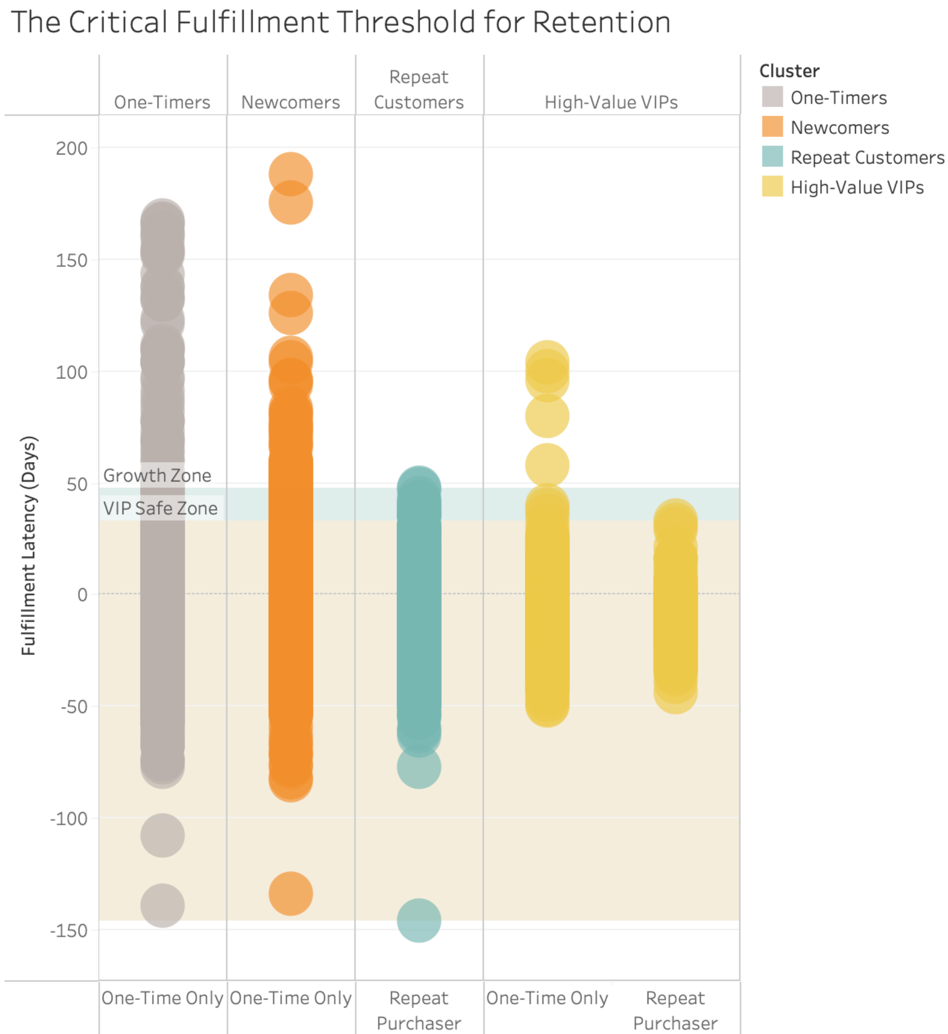
Objective

Translate delivery-delay risk into an operational warning system that can support proactive CRM intervention.

Rationale

The segmented Logistic Regression results show that delivery delays reduce repurchase likelihood, especially among premium customers. NLP review mining also shows that delivery-related terms appear frequently in customer reviews, suggesting that fulfillment experience is a visible part of customer satisfaction.

To make this finding actionable, fulfillment latency was compared across one-time and repeat customers within each customer group.



The chart compares fulfillment latency between one-time purchasers and repeat purchasers across customer groups.

Repeat purchasers tend to remain within lower fulfillment-latency ranges. Based on the observed distribution, delays beyond approximately 48 days for the general customer base and around 33 days for High-Value VIPs can be treated as exploratory warning zones.

These thresholds should not be interpreted as strict causal cutoffs. Instead, they should be used as operational benchmarks to identify orders that may require proactive service recovery.

Business Interpretation

The fulfillment-latency view turns the delivery-delay finding into an actionable CRM mechanism. When an order approaches the warning zone, Olist can intervene before the customer experience deteriorates further.

This framework provides the operational foundation for the retention strategies in the next section.

8. Strategic Recommendations

Based on the analysis, Olist should move from one-size-fits-all retention management to segment-specific CRM actions based on customer value, price sensitivity, and fulfillment risk.

Strategy	Objective	Rationale	Recommended Actions	Success Metric
8.1 Value Segment Strategy	Increase repurchase through cost reduction and repeat-purchase activation.	Decision Tree and Logistic Regression results suggest that lower-order-value customers are more sensitive to product price and freight cost. For this group, shipping cost can create meaningful purchase friction.	• Free shipping A/B test • Shipping subsidies • bundled promotions • First repeat-purchase coupons	90-day repurchase rate
8.2 Premium Segment Strategy	Increase retention through service quality and fulfillment reliability.	Segmented Logistic Regression shows that premium customers are less sensitive to price and freight cost, but more sensitive to delivery delays. The fulfillment early-warning framework further suggests that premium customers should be monitored before delivery latency reaches the risk zone.	• Priority fulfillment • VIP shipping program • Delivery ETA accuracy • Premium membership benefits • Proactive service recovery for delayed orders	VIP retention rate
8.3 Delayed-Order Recovery Strategy	Recover customers whose orders are approaching fulfillment-risk warning zones.	The early-warning framework identifies delayed orders before they turn into churn risk. This allows Olist to respond before customers decide not to return.	• Delivery status updates • Apology messages • Customer service escalation • Shipping fee credits • VIP priority support	Repurchase rate among delayed-order customers

Project Assets

Technical repository includes:

Asset	Included Materials
SQL Scripts	Data cleaning; repurchase rate calculation; RFM feature engineering
Python Notebooks	Data preprocessing; K-Means clustering; Decision Tree threshold analysis; segmented Logistic Regression; NLP review mining
Tableau Workbook / Dashboard Link	Interactive marketplace performance dashboard; customer segmentation views; retention and fulfillment-risk analysis views
Methodology Notes	Feature definitions; modeling assumptions; business interpretation

GitHub Link: https://github.com/chloeliu-analytics/olist-fashion-retention-analysis